The space-time hardness of lattice problems

Martin R. Albrecht

ICQCS'26 (20 March 2026)

Hardness of hinted ISIS from the space-time hardness of lattice problems

joint work with Russell W. F. Lai and Eamonn W. Postlethwaite

https://ia.cr/2026/187

Motivation for those who like applications: Don't sign it twice!

Mike Ounsworth 18 Sept 2022, 20:42:53

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Christopher J Peikert 19 Sept 2022, 03:45:26

to Mike Ounsworth, pqc-forum, p...@ietf.org, cf...@irtf.org

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For Falcon in particular, there is a much more essential reason for "keying" the hash by a random number: releasing two different signatures for the same hash digest breaks the security reduction, and likely has very bad practical consequences as well. Note that the "core" signing procedure -- given a hash digest, output a signature -- is randomized, so different runs on the same digest will tend to produce different signatures.

Randomized hashing ensures that the same digest essentially never reoccurs (assuming quality randomness), even when signing the same message multiple times. See Section 2.2.2 of the Falcon spec for further details.

https://groups.google.com/a/list.nist.gov/g/pqc-forum/c/cIsc6tUY9Rw/m/E0fPG7QkAQAJ

verification key

$\mathbf{A} \in \mathbb{Z}_q^{n \times m}$

signing key

short $\mathbf{T}$ s.t. $\mathbf{A} \cdot \mathbf{T} \equiv \mathbf{0} \bmod q$

signature

$(\mathbf{u}, r)$ s.t. $H(m, r) \equiv \mathbf{A} \cdot \mathbf{u} \bmod q$

with $\mathbf{u}$ short

Motivation for those who like lattices: An enticing lie

Efficient Adversaries

Some lattice problems and their relations

Definition (γ-SIVP)

Given $\gamma \geq 1$ and a full-rank lattice $\Lambda \subset \mathbb{R}^n$, find linearly independent $\mathbf{v}_1, \ldots \mathbf{v}_n$ such that $\max \|\mathbf{v}_i\| \leq \gamma \cdot \lambda_n(\Lambda)$, where $\lambda_n(\Lambda)$ is the n-th successive minimum of Λ.

Definition (SIS ([Ajt96]))

Given a random $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$, return $\mathbf{u} \in \mathbb{Z}^m$ with $0 < \|\mathbf{u}\| \leq \beta$ such that $\mathbf{A} \cdot \mathbf{u} \equiv \mathbf{0} \bmod q$.

Theorem (SIVP $\implies$ SIS [Ajt96; MR07])

Let $m, \beta \in \mathrm{poly}(n)$, $q \geq 8\, n \cdot \sqrt{m} \cdot \beta$ and $\gamma = 8\, \beta \cdot \sqrt{n} \cdot \omega(\sqrt{\log n})$. Then solving SIS for n, m, q, β in time τ and memory μ, solves γ-SIVP in time $\tau + \mathrm{poly}(n)$ and memory $\mu + \mathrm{poly}(n)$.

Notation

Write

- $\mathrm{SIS}_{n,m,q,\beta}$ to represent the SIS problem.
- (τ, μ)-$\mathrm{SIS}_{n,m,q,\beta}$ for the set of algorithms that solve $\mathrm{SIS}_{n,m,q,\beta}$ in time τ, memory μ and with non-negligible success probability.
- $\Lambda_q^{\perp}(\mathbf{A}) := \{\mathbf{x} \in \mathbb{Z}^m : \mathbf{A} \cdot \mathbf{x} \equiv \mathbf{0} \bmod q\}$ for the kernel lattice of $\mathbf{A}$. SIS asks to find short elements in this lattice.
- $D_{\Lambda,s,\mathbf{c}}$ for the discrete Gaussian distribution with support Λ, parameter s and centre $\mathbf{c}$.

"Let $\mathcal{A}$ be a ~~PPT~~ BQP adversary ..."

Assumption (SIS)

For sensible $m, q, \beta \in \mathrm{poly}(n)$ *we have* $(\mathrm{poly}(n), \mathrm{poly}(n))$-$\mathrm{SIS}_{n,m,q,\beta} = \emptyset$.

"Let $\mathcal{A}$ be a subexponential adversary ..."

Assumption (Subexponential-secure SIS)

For sensible $m, q, \beta \in \mathrm{poly}(n)$ *we have* $\left(2^{o(n)}, 2^{o(n)}\right)$ -$\mathrm{SIS}_{n,m,q,\beta} = \emptyset$.

Algorithms for finding short vectors / solving SIS

Enumeration

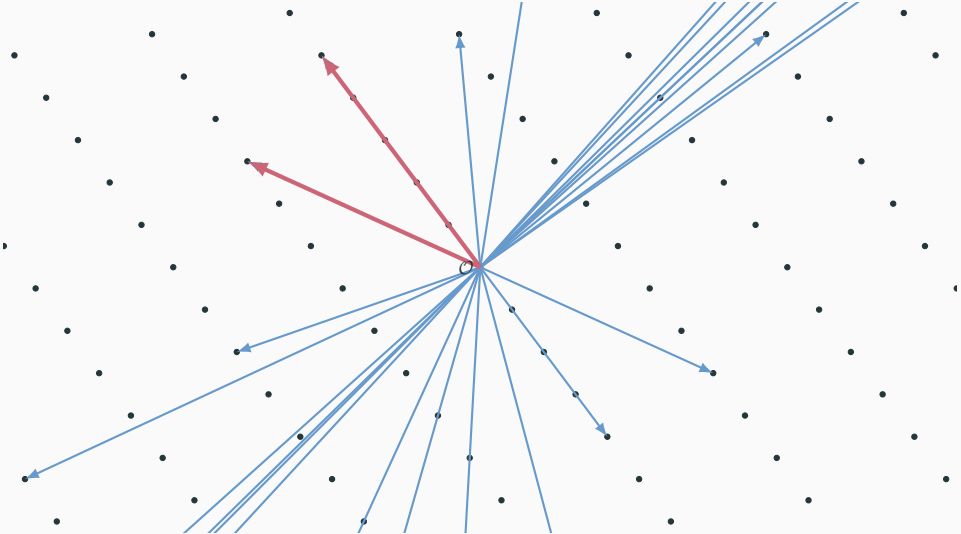
- Search through vectors smaller than a given bound: project down to 1-dim problem, lift to 2-dim problem ...
- **Time:** $n^{\Theta(n)}$
- **Memory:** $\mathrm{poly}(n)$

Sieving

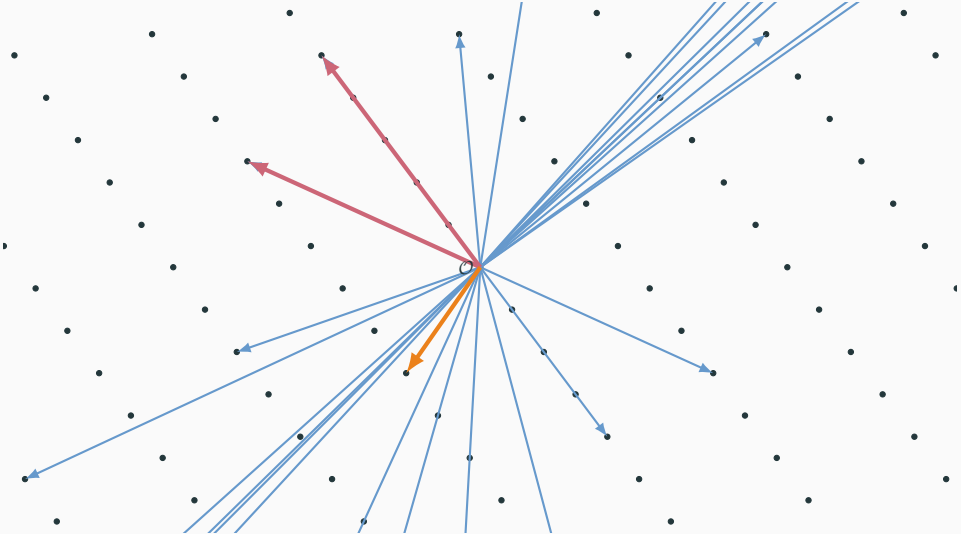
- Produce new, shorter vectors by considering sums and differences of existing vectors
- **Time:** $2^{\Theta(n)}$
- **Memory:** $2^{\Theta(n)}$

Sieving I

Sieving II



Sieving III



Space-time hardness: cryptanalysis

- "It is still unclear if we can get a $2^{O(n)}$ algorithm that uses only polynomial space"[1]
- "is it possible to achieve exponential time complexity with a polynomially bounded space requirement?"[2]
- "A more general open problem is whether SVP can be solved in singly exponential time but only polynomial space."[3]

---

[1] Daniele Micciancio and Panagiotis Voulgaris. **Faster Exponential Time Algorithms for the Shortest Vector Problem**. In: *21st SODA*. ed. by Moses Charika. ACM-SIAM, Jan. 2010, pp. 1468–1480. DOI: 10.1137/1.9781611973075.119.

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[3] Divesh Aggarwal, Daniel Dadush, Oded Regev and Noah Stephens-Davidowitz. **Solving the Shortest Vector Problem in 2^n Time Using Discrete Gaussian Sampling: Extended Abstract**. In: *47th ACM STOC*. ed. by Rocco A. Servedio and Ronitt Rubinfeld. ACM Press, June 2015, pp. 733–742. DOI: 10.1145/2746539.2746606.

Space-time hardness: conjecture

Corollary 2.2. *The conclusion of Theorem 2.2 follows from the assumption that worst-case* $\gamma(n)$-GapSVP *(for a fixed* $\gamma(n) = \mathrm{poly}(n)$*) cannot be solved in time* $n^{o(n)}$ *with* $\mathrm{poly}(n)$ *space and* $\mathrm{poly}(n)$ *bits of nonuniform advice (independent of the lattice).*

Polynomial-space algorithms for GapSVP have themselves been an object of study for over 25 years [Kan83, KF16, BLS16, ABF+20], but the current best (poly-space) algorithms for this problem run in time $n^{\Omega(\epsilon n)}$ for approximation factor $n^{1/\epsilon}$. Therefore, under a sufficiently strong (and plausible) worst-case assumption about GapSVP, we have a polynomial-time Fiat-Shamir compiler without complexity leveraging.

Alex Lombardi and Vinod Vaikuntanathan. Fiat-Shamir for Repeated Squaring with Applications to PPAD-Hardness and VDFs. In: *CRYPTO 2020, Part III*. ed. by Daniele Micciancio and Thomas Ristenpart. Vol. 12172. LNCS. Springer, Cham, Aug. 2020, pp. 632–651. DOI: 10.1007/978-3-030-56877-1_22

A COROLLARY

Assumption (Space-time hard SIS)

For sensible $m, q, \beta \in \mathrm{poly}(n)$ *we have* $\left(2^{O(n)}, \mathrm{poly}(n)\right)$ -$\mathrm{SIS}_{n,m,q,\beta} = \emptyset$.

A COROLLARY (A SPECIALISATION)

Assumption (Space-time hard SIS)

For sensible $q, \beta \in \mathrm{poly}(n)$, $m \in O(n)$ *we have* $\left(2^{O(m)}, \mathrm{poly}(n)\right)$ -$\mathrm{SIS}_{n,m,q,\beta} = \emptyset$.

- When $m \leq O(n)$ then $2^{O(n)} = 2^{O(m)}$; we consider only this case in this talk.
- In the paper we also consider $m \leq o(n \log n)$, which is covered by the conjecture.
- When $m \geq \Omega(n \log n)$, there are algorithms in $\left(2^{O(m)}, \mathrm{poly}(n)\right)$ -$\mathrm{SIS}_{n,m,q,\beta}$.

Hinted lattice problems

H-SIS

$\mathsf{Exp\text{-}H\text{-}ISIS}_{(k,m,q,\beta,s),\mathcal{A}}(1^n)$

$(\mathbf{A}, \mathbf{t}) \leftarrow \mathbb{Z}_q^{n \times m} \times \mathbb{Z}_q^n$

$(\mathbf{u}_1, \ldots, \mathbf{u}_k) \leftarrow D_{\Lambda_q^\perp(\mathbf{A}),s,\mathbf{0}}$ // $\|\mathbf{u}_i\| \approx s\sqrt{m}$

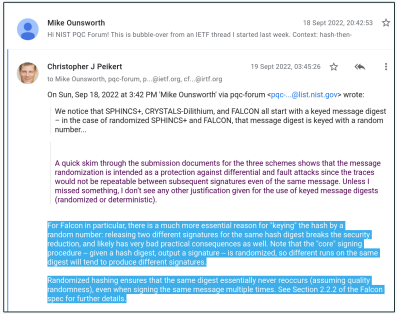
$\mathbf{u} \leftarrow \mathcal{A}(\mathbf{A}, \mathbf{t}, \mathbf{u}_1, \ldots, \mathbf{u}_k)$

return $[\![\mathbf{A} \cdot \mathbf{u} = \mathbf{t} \wedge \|\mathbf{u}\| \leq \beta]\!]$

- $\beta \geq \Omega(\sqrt{m} \max \|\mathbf{u}_i\|)$: obviously easy: run Babai's Nearest Plane algorithm
- $\beta \in O(1) \cdot \min \|\mathbf{u}_i\|$: plausibly hard[a]

[a] Shweta Agrawal, Elena Kirshanova, Damien Stehlé and Anshu Yadav. Practical, Round-Optimal Lattice-Based Blind Signatures. In: *ACM CCS 2022*. Ed. by Heng Yin, Angelos Stavrou, Cas Cremers and Elaine Shi. ACM Press, Nov. 2022, pp. 39–53. DOI: 10.1145/3548606.3560650.

Relation to motivations



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Randomized hashing ensures that the same digest essentially never reoccurs (assuming quality randomness), even when signing the same message multiple times. See Section 2.2.2 of the Falcon spec for further details.

Two signatures $\mathbf{v}_1, \mathbf{v}_2$ for the same (m, r) in Falcon, give $\mathbf{v}_1 - \mathbf{v}_2$ s.t. $\mathbf{v}_1 - \mathbf{v}_2 \sim D_{\Lambda_q^{\perp}(\mathbf{A}), \sqrt{2}s, \mathbf{0}}$. Forgery means solving ISIS with similar norm.

This is precisely the H-ISIS problem.

Reduction

Main result

Theorem

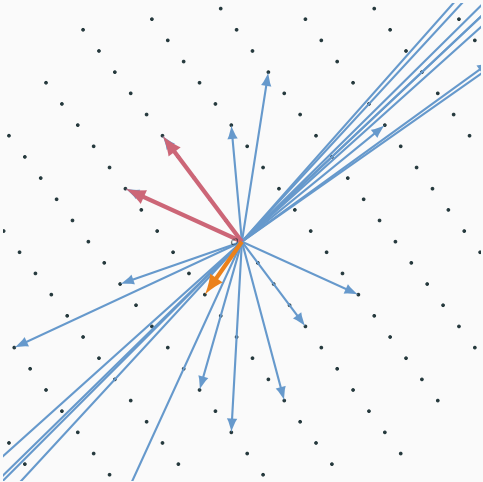
$(2^{O(n)}, \mathrm{poly}(n))$ -poly(n)-$SIVP = \emptyset$,
$\implies$ There exists k, β, s such that $(\mathrm{poly}(n), \mathrm{poly}(n))$ -H-ISIS$_{k,n,O(n),\mathrm{poly}(n),\beta,s} = \emptyset$.

Geometry of sieves: "There cannot be too many points in a ball that are far apart."

Lemma

Let $S \subset \mathrm{B}^m(\beta)$.
If $|S| > 2^{m \log(1+2\gamma)}$, *there exist distinct* $\mathbf{c}', \mathbf{c}'' \in S$ *with* $\|\mathbf{c}' - \mathbf{c}''\| \leq \beta/\gamma$.

Remark: Cost estimates for lattice schemes such as Falcon or Kyber use a heuristic variant of this theorem.



Probabilistic upgrade

Lemma ("No distribution over a ball can avoid close pairs.")

*For **any (malicious) distribution** D over $\mathrm{B}^m(\beta)$ with finite support:*

$$\Pr_{\mathbf{c}',\mathbf{c}'' \leftarrow D}[\|\mathbf{c}' - \mathbf{c}''\| \leq \beta/\gamma] \geq \frac{1}{2^{m \log(1+2\gamma)}}.$$

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Theorem (Motzkin–Straus Theorem [MS65])

Let G be an undirected simple graph with adjacency matrix $\mathbf{M}$.

$$\max_{\mathbf{d}\in[0,1]^n,\|\mathbf{d}\|_1=1} \mathbf{d}^T\cdot\mathbf{M}\cdot\mathbf{d} = 1-\frac{1}{\omega(G)}$$

where $\omega(G) = \max\{|X| : X\subseteq V \wedge (\forall(x,y\in X\wedge x\neq y),\ \{x,y\}\in E)\}$ is the clique number.

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Define G so that $\mathbf{c}$ and $\mathbf{c}'$ are connected iff $\|\mathbf{c}'-\mathbf{c}''\| > \beta/\gamma$. Clique = "(anti-)cluster" of far-apart points. Apply Motzkin–Straus and geometry of sieves: $\omega(G) \leq 2^{m\log(1+2\gamma)}$.

A sieving algorithm

$\mathcal{B}^{\mathcal{A}}(\mathsf{A}, \{\mathsf{u}_i\}_{1 \le i \le k})$

```
s := β/(2γ√(2m))
for up to 2^{O(m·(1+log 4γ))} times
    v_ℓ, v_r ← D_{Z^m,s} × D_{Z^m,s}
    v'_ℓ ← A(A, A·v_ℓ, {u_i}_{1≤i≤k})
    v'_r ← A(A, A·v_r, {u_i}_{1≤i≤k})
    if ‖v'_ℓ − v'_r‖ ≤ β/(2γ)
        return (v_ℓ − v_r) − (v'_ℓ − v'_r)
```

A sieving algorithm

$\mathcal{B}^{\mathcal{A}}(\mathbf{A}, \{\mathbf{u}_i\}_{1 \le i \le k})$

$s := \beta/(2\gamma\sqrt{2m})$

for up to $2^{O(m\cdot(1+\log 4\gamma))}$ times

  $\mathbf{v}_\ell, \mathbf{v}_r \leftarrow D_{\mathbb{Z}^m,s} \times D_{\mathbb{Z}^m,s}$

  $\mathbf{v}'_\ell \leftarrow \mathcal{A}(\mathbf{A}, \mathbf{A}\cdot\mathbf{v}_\ell, \{\mathbf{u}_i\}_{1 \le i \le k})$

  $\mathbf{v}'_r \leftarrow \mathcal{A}(\mathbf{A}, \mathbf{A}\cdot\mathbf{v}_r, \{\mathbf{u}_i\}_{1 \le i \le k})$

  if $\|\mathbf{v}'_\ell - \mathbf{v}'_r\| \le \beta/(2\gamma)$

    return $(\mathbf{v}_\ell - \mathbf{v}_r) - (\mathbf{v}'_\ell - \mathbf{v}'_r)$

- $\gamma \in O(1) \implies 2^{O(m\cdot(1+4\gamma))} = 2^{O(m)}$
- $\mathbf{0} \equiv \mathbf{A}\cdot(\mathbf{v}_i - \mathbf{v}'_i) \bmod q$ for $i \in \{\ell, r\}$
  $\implies \mathbf{0} \equiv \mathbf{A}\cdot(\mathbf{v}_\ell - \mathbf{v}_r) - (\mathbf{v}'_\ell - \mathbf{v}'_r) \bmod q$
- $(\mathbf{v}_\ell - \mathbf{v}_r) - (\mathbf{v}'_\ell - \mathbf{v}'_r) \approx_s D_{\Lambda_q^\perp(\mathbf{A}), \sqrt{2}s, \mathbf{v}'_r - \mathbf{v}'_\ell}$

Cleaning up the output distribution

An adaptation of [DFPS22] gives us:

Have $Q'_{(i,j)} := D_{\Lambda_q^\perp(\mathbf{A}),s,\mathbf{v}'_i} - D_{\Lambda_q^\perp(\mathbf{A}),s,\mathbf{v}'_j}$

Close to $Q_{(i,j)} := D_{\Lambda_q^\perp(\mathbf{A}),\sqrt{2}s,\mathbf{v}'_i-\mathbf{v}'_j}$ with $\Delta(Q'_{(i,j)}, Q_{(i,j)}) \leq 2^{-\Omega(m)}$

Want $P_i := D_{\Lambda_q^\perp(\mathbf{A}),\sqrt{2}s,0}$

Get P'_i with $\Delta(P'_i, P_i) \leq 2^{-\Omega(m)}$ in $(\tau, \mu) = (2^{O(m)}, \mathrm{poly}(n))$

Punchline

We can "clean up" the distribution output by our sieve to be close to a zero-centred Gaussian like the hints but a constant factor γ shorter.

Recursion

Assume there is a chain of such $\{\mathcal{B}_i\}_{1 \leq i \leq z}$.

- Feed the outputs of $\mathcal{B}_i$ to $\mathcal{B}_{i+1}$ as hints.
- This allows us to make progress $\gamma^z \geq 2^{\Omega(\mathrm{poly}(n))}$.
- Start the chain by sampling from a discrete Gaussian with $s_1 \in \omega(\log m) \cdot \sqrt{m} \cdot q$ – which is easy – to produce the required initial hints.

Stopping condition

At some point, this process has to stop working, because there simply cannot be some $\mathcal{B}_z$ for the required parameters, in particular β_z.

- We establish that when
$$\beta_z \geq \Omega(m) \cdot \lambda_m(\Lambda_q^{\perp}(\mathbf{A}))$$
then a polynomially number of Gaussians sampled by $\mathcal{B}_z$ will span the entire lattice with overwhelming probability.
- We also establish that
$$\lambda_m(\Lambda_q^{\perp}(\mathbf{A})) \in O(\sqrt{\log m}) \cdot \lambda_1(\Lambda_q^{\perp}(\mathbf{A})),$$
suggesting that such a best-case $\mathcal{A}_z$ would sample solutions within a factor of $O(m \cdot \sqrt{\log m})$ from optimal: $\lambda_1(\Lambda_q^{\perp}(\mathbf{A}))$.

Caveats

- In general, we cannot derive the existence of $\mathcal{B}_{i+1}$ from $\mathcal{B}_i$
- It is not a given that $\mathcal{B}_i$ and $\mathcal{B}_{i+1}$ "agree" on which **A** they accept as inputs, i.e. succeed on them.

Some H-ISIS must be hard if lattices are space-time hard

For a given $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$:

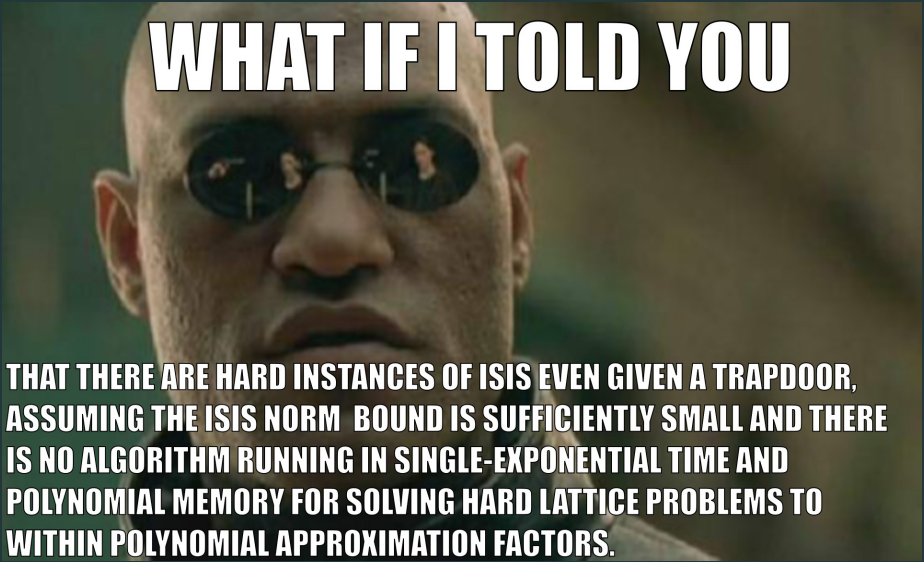
Either The H-ISIS instances for all parameters $\{(m, n, q, \beta_i, s_i)\}_{1 \leq i \leq z}$ are easy,

- which implies SIS for $\mathbf{A}$ is space-time easy,
- which means worst-case lattice problems are space-time easy

Or H-ISIS is hard for at least one set of the parameters (m, n, q, β_j, s_j)

- but our reduction does not establish which j, i.e. we do not formally establish that the problem does not get easier as norm bounds shrink.

Fin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THAT THERE ARE HARD INSTANCES OF ISIS EVEN GIVEN A TRAPDOOR, ASSUMING THE ISIS NORM BOUND IS SUFFICIENTLY SMALL AND THERE IS NO ALGORITHM RUNNING IN SINGLE-EXPONENTIAL TIME AND POLYNOMIAL MEMORY FOR SOLVING HARD LATTICE PROBLEMS TO WITHIN POLYNOMIAL APPROXIMATION FACTORS.

References I

[ADRS15] Divesh Aggarwal, Daniel Dadush, Oded Regev and Noah Stephens-Davidowitz. **Solving the Shortest Vector Problem in 2^n Time Using Discrete Gaussian Sampling: Extended Abstract.** In: *47th ACM STOC*. Ed. by Rocco A. Servedio and Ronitt Rubinfeld. ACM Press, June 2015, pp. 733–742. DOI: 10.1145/2746539.2746606.

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[LV20] Alex Lombardi and Vinod Vaikuntanathan. **Fiat-Shamir for Repeated Squaring with Applications to PPAD-Hardness and VDFs.** In: *CRYPTO 2020, Part III*. Ed. by Daniele Micciancio and Thomas Ristenpart. Vol. 12172. LNCS. Springer, Cham, Aug. 2020, pp. 632–651. DOI: 10.1007/978-3-030-56877-1_22.

[MR07] Daniele Micciancio and Oded Regev. **Worst-Case to Average-Case Reductions Based on Gaussian Measures.** In: *SIAM Journal on Computing* 37.1 (2007), pp. 267–302. DOI: 10.1137/S0097539705447360. eprint: https://doi.org/10.1137/S0097539705447360. URL: https://doi.org/10.1137/S0097539705447360.

[MS65] T. S. Motzkin and E. G. Straus. **Maxima for Graphs and a New Proof of a Theorem of Turán.** In: *Canadian Journal of Mathematics* 17 (1965), pp. 533–540. DOI: 10.4153/CJM-1965-053-6.

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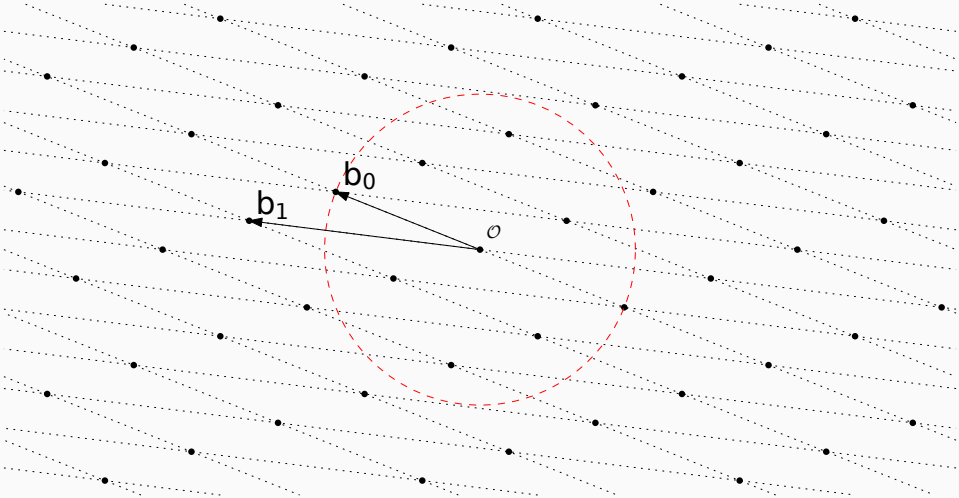
[MV10] Daniele Micciancio and Panagiotis Voulgaris. **Faster Exponential Time Algorithms for the Shortest Vector Problem**. In: *21st SODA*. Ed. by Moses Charika. ACM-SIAM, Jan. 2010, pp. 1468–1480. DOI: 10.1137/1.9781611973075.119.

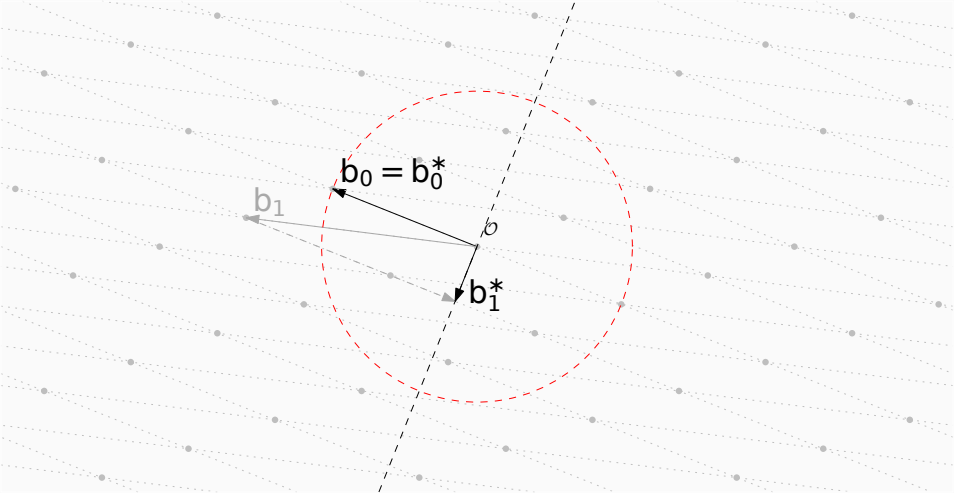
Enumeration

Enumeration I – Pick a radius



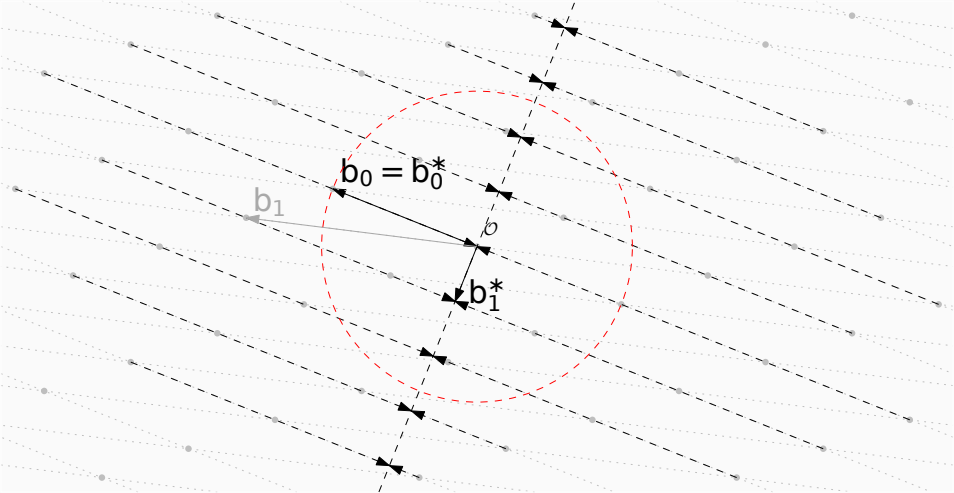
Picture credit: Joop van de Pol

Enumeration II – Project basis



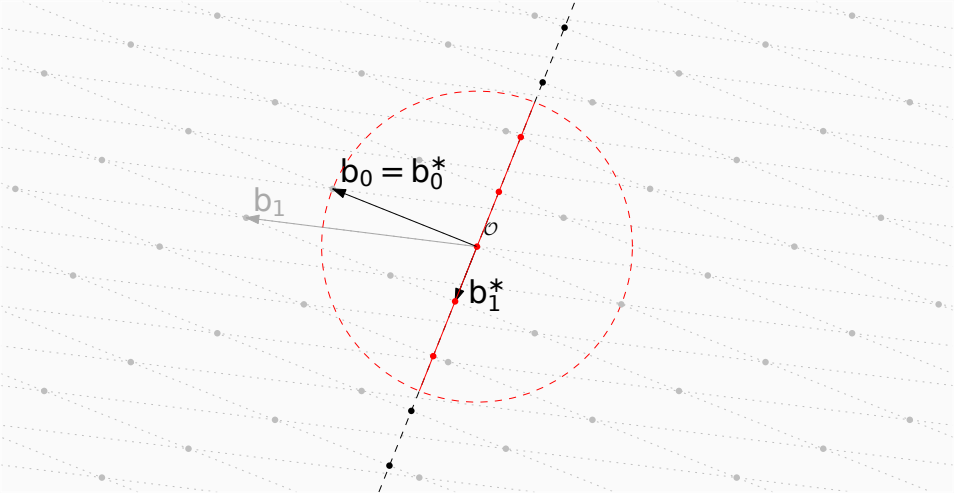
Picture credit: Joop van de Pol

Enumeration III – Project lattice



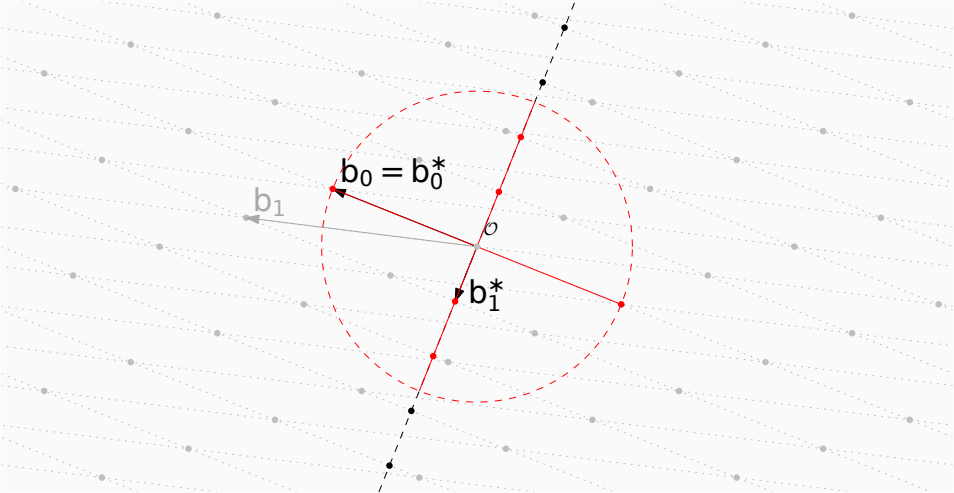
Picture credit: Joop van de Pol

Enumeration IV – Enumerate projections



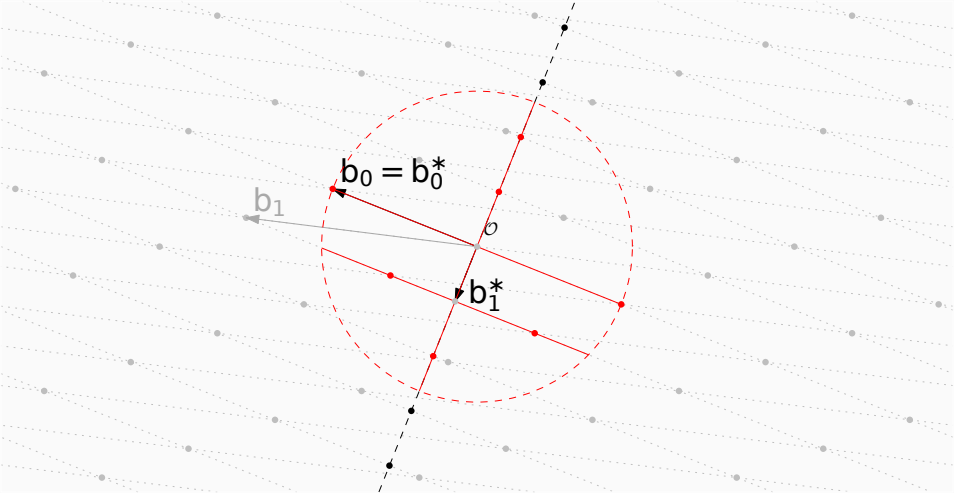
Picture credit: Joop van de Pol

Enumeration V – For each lift and enumerate



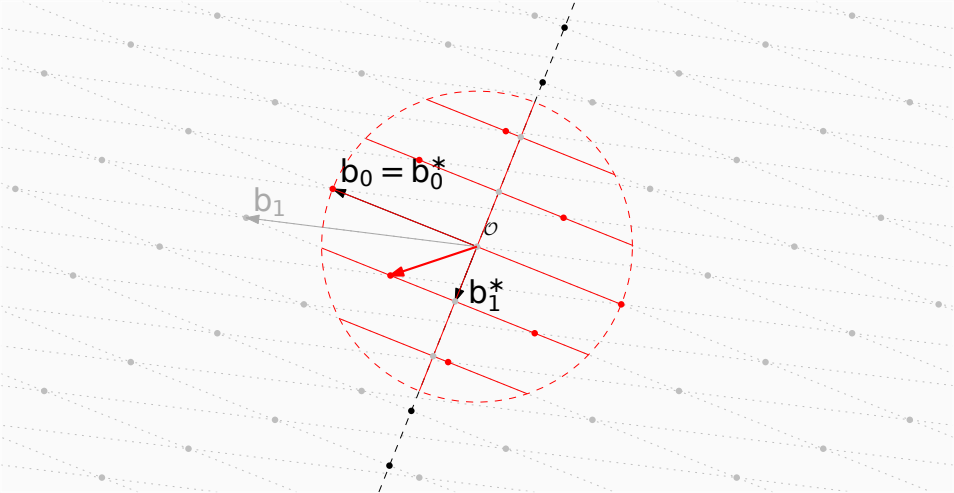
Picture credit: Joop van de Pol

Enumeration V – For each lift and enumerate



Picture credit: Joop van de Pol

Enumeration VI – Keep shortest



Picture credit: Joop van de Pol